

The Exponent Rules (Part III) Homework

Simplify each expression below and write the Roman numerals of the exponent rules that you use beside each step. Do not leave your answers with fraction exponents or negative exponents.

$$\textcircled{1} \left(\frac{x^3 \cdot c^4}{x^3} \right)^{-3}$$

$$\textcircled{8} \left(\frac{y^6 \cdot b^7}{b^9} \right)^{-5}$$

$$\textcircled{2} (x^2 \cdot x^5 \cdot y^8)^{\frac{1}{7}}$$

$$\textcircled{9} \left(\frac{x^{\frac{1}{3}} \cdot y^2}{b^{10}} \right)^3$$

$$\textcircled{3} \left(\frac{s^{-1}}{r^{-\frac{9}{8}}} \right)^{-\frac{8}{9}}$$

$$\textcircled{10} 5 \cdot 7^0 \cdot 5^{-3}$$

$$\textcircled{4} (x \cdot x^2)^{-3}$$

$$\textcircled{11} \left(\frac{x^3 \cdot 8^5}{8^4} \right)^2$$

$$\textcircled{5} \left(\frac{a^5 \cdot a^4}{a^2} \right)^{-7}$$

$$\textcircled{12} (64^{-3} \cdot 64^5)^{-\frac{1}{6}}$$

$$\textcircled{6} \left(\frac{y^5}{x^8 \cdot x^3} \right)^{\frac{2}{11}}$$

$$\textcircled{7} \left(\frac{b^3 \cdot b^2}{a^4 \cdot c^0} \right)^4$$

$$\textcircled{13} \frac{1}{b^{-5}}$$

$$\textcircled{14} \frac{1}{8^{-\frac{1}{3}}}$$

$$\textcircled{15} \frac{-7}{x^{-4}}$$

$$\textcircled{16} \frac{10y^{17}}{b^{-3}}$$

$$\textcircled{17} \frac{4a^8}{3^{-2}}$$

$$\textcircled{18} \frac{7x^5y^3}{a^{-2}b^{-10}}$$

$$\textcircled{19} \frac{1}{25^{-\frac{3}{2}}}$$

$$\textcircled{20} \frac{6a^9}{(-2)^{-3}}$$

$$\textcircled{21} \frac{x^4}{4b^{-7}}$$

$$\textcircled{22} \frac{8x^5}{11a^3y^{-2}}$$

$$\textcircled{23} \frac{a^{10}}{4^{-3}x^5}$$

$$\textcircled{24} \frac{x^5y^{-3}}{a^{-8}b^7}$$

$$\textcircled{25} \frac{54c^4b^{-5}}{9x^4y^{-2}}$$

$$(26) \frac{8x^{-9}a^3}{40y^{-10}}$$

$$(27) \frac{15a^{-4}b^{-8}}{25c^{-3}}$$

$$(28) \frac{72x^{-1}}{56y^{-1}}$$

$$(29) \frac{100x^{14}}{10y^{-6}}$$

$$(30) \frac{2a^{-4}b^{-7}}{10x^2y^{11}}$$

$$(31) \left(\frac{3^{-2}}{x^4}\right)^2$$

$$(32) \left(\frac{2}{y^{-8}}\right)^3$$

$$(33) \left(\frac{b^{-3}}{7}\right)^2$$

$$(34) \left(\frac{3}{4}\right)^{-2}$$

$$(35) \left(\frac{1}{5}\right)^{-3}$$

$$(36) \left(\frac{2}{7}\right)^{-1}$$

$$(37) (\sqrt{4^2 \cdot 4^3})^{-1}$$

$$(38) \left(\sqrt[3]{\frac{8^7}{8^3}}\right)^{-\frac{1}{2}}$$

$$(39) (\sqrt[4]{81^2 \cdot 81^3})^{-1}$$

Answers

$$\text{II, VII, V} \quad \text{VIII} \quad \textcircled{1} \quad \frac{1}{c^{12}}$$

$$\text{II, III, V, VIII} \quad \textcircled{8} \quad \frac{b^{10}}{y^{30}}$$

$$\text{I, III, VI} \quad \text{X} \quad \textcircled{2} \quad x(\sqrt[7]{y})^8$$

$$\text{IV, III, V, VI} \quad \textcircled{9} \quad \frac{xy^6}{b^{30}}$$

$$\text{IV, V, VI} \quad \text{X} \quad \textcircled{3} \quad \frac{(\sqrt[9]{s})^8}{r}$$

$$\text{VII, VI, I, VIII} \quad \textcircled{10} \quad \frac{1}{25}$$

$$\text{VI, I, V} \quad \text{VIII} \quad \textcircled{4} \quad \frac{1}{x^9}$$

$$\text{II, VI, III, I} \quad \textcircled{11} \quad 64x^6$$

$$\text{I, II, V} \quad \text{VIII} \quad \textcircled{5} \quad \frac{1}{a^{49}}$$

$$\text{I, V, VIII, IX} \quad \textcircled{12} \quad \frac{1}{4}$$

$$\text{I, IV, V} \quad \text{X} \quad \textcircled{6} \quad \frac{(\sqrt[11]{y})^{10}}{x^2}$$

$$\text{VIII}' \quad \textcircled{13} \quad b^5$$

$$\text{VIII}', \text{IX} \quad \textcircled{14} \quad 2$$

$$\text{I, VII, IV} \quad \text{V} \quad \textcircled{7} \quad \frac{b^{20}}{a^{16}}$$

$$\text{VIII}' \quad \textcircled{15} \quad -7x^4$$

$$\text{VIII}' \quad \textcircled{16} \quad 10b^3y^{17}$$

$$\text{VIII}' \quad \textcircled{23} \quad \frac{64a^{10}}{x^5}$$

$$\text{VIII}' \quad \textcircled{17} \quad 36a^8$$

$$\text{VIII, VIII}' \quad \textcircled{24} \quad \frac{x^5a^8}{b^7y^3}$$

$$\text{VIII}' \quad \textcircled{18} \quad 7x^5y^3a^2b^{10}$$

$$\text{VIII, VIII}' \quad \textcircled{25} \quad \frac{6c^4y^2}{x^4b^5}$$

$$\text{VIII}', \text{X} \quad \textcircled{19} \quad 125$$

$$\text{VIII, VIII}' \quad \textcircled{26} \quad \frac{a^3y^{10}}{5x^9}$$

$$\text{VIII}' \quad \textcircled{20} \quad -48a^9$$

$$\text{VIII, VIII}' \quad \textcircled{27} \quad \frac{3c^3}{5a^4b^8}$$

$$\text{VIII}' \quad \textcircled{21} \quad \frac{x^4b^7}{4}$$

$$\text{VIII, VIII}' \quad \text{VI} \quad \textcircled{28} \quad \frac{9y}{7x}$$

$$\text{VIII}' \quad \textcircled{22} \quad \frac{8x^5y^2}{11a^3}$$

$$\text{VIII}' \quad \textcircled{29} \quad 10x^{14}y^6$$

$$\text{VIII } (30) \begin{array}{|c|c|} \hline & \\ \hline 5x^2y^{11}a^4b^7 & \\ \hline \end{array}$$

$$\text{I, X', VIII, VI } (37) \begin{array}{|c|} \hline 1 \\ \hline 32 \\ \hline \end{array}$$

Other paths are possible.

$$\begin{array}{l} \text{VIII, IV} \\ \text{III, V} \end{array} (31) \begin{array}{|c|} \hline 1 \\ \hline 81x^8 \\ \hline \end{array}$$

$$\text{II, X', VIII, IX } (38) \begin{array}{|c|} \hline 1 \\ \hline 4 \\ \hline \end{array}$$

$$\begin{array}{l} \text{VIII', III} \\ \text{V} \end{array} (32) \begin{array}{|c|} \hline 8y^{24} \\ \hline \end{array}$$

$$\text{I, X', VIII, VI } (39) \begin{array}{|c|} \hline 1 \\ \hline 243 \\ \hline \end{array}$$

$$\begin{array}{l} \text{VIII, IV} \\ \text{III, V} \end{array} (33) \begin{array}{|c|} \hline 1 \\ \hline 49b^6 \\ \hline \end{array}$$

$$\text{VIII } (34) \begin{array}{|c|} \hline 16 \\ \hline 9 \\ \hline \end{array}$$

$$\text{VIII } (35) \begin{array}{|c|} \hline 125 \\ \hline \end{array}$$

$$\text{VIII, VI } (36) \begin{array}{|c|} \hline 7 \\ \hline 2 \\ \hline \end{array}$$